RED SIFT

# Monitoring Brand Logos through Certificate Transparency

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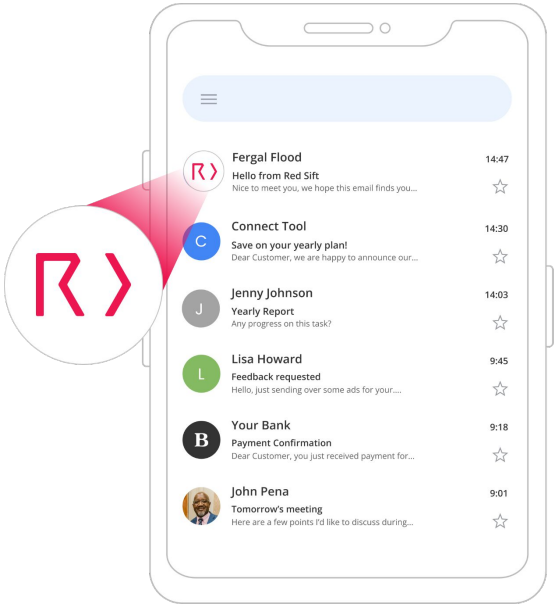
# Agenda

1. What is BIMI and how it works?
2. Email Clients
3. Certifying authorities
4. Certificate transparency
5. Insights from CT monitoring

# BIMI

Enables businesses to show their brand logo next to every email they send




Brand Indicators for Message Identification (BIMI) is an email standard that enables organizations to display their verified brand logo alongside DMARC-authenticated emails, with broad adoption beginning in 2021.

## DMARC

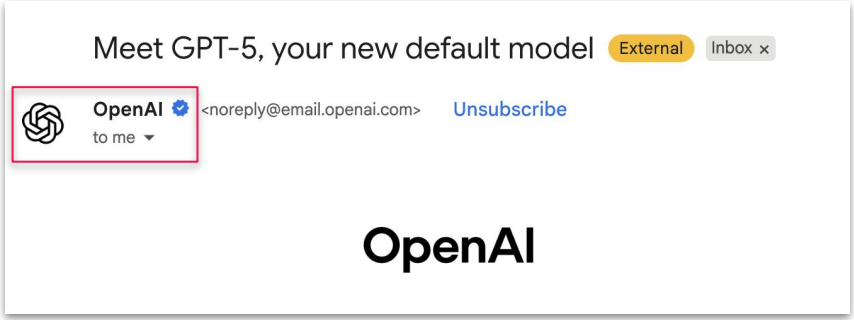
**Domain-based Message Authentication, Reporting, and Conformance** builds on SPF and DKIM to verify sender authenticity and alignment, instructing mail servers to reject or quarantine emails that fail authentication and help prevent domain spoofing.

## SPF & DKIM

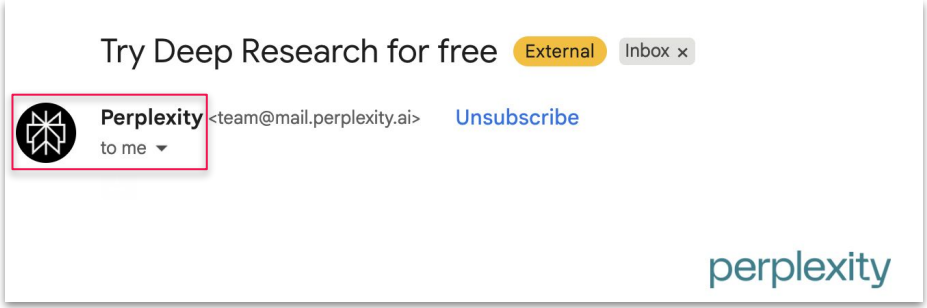
**Sender Policy Framework (SPF)** verifies that an email comes from an authorized IP address listed in the domain's DNS SPF record

**DomainKeys Identified Mail** (DKIM) ensures message integrity and authenticity by attaching a cryptographic signature to the email header, generated using the sender's private key.

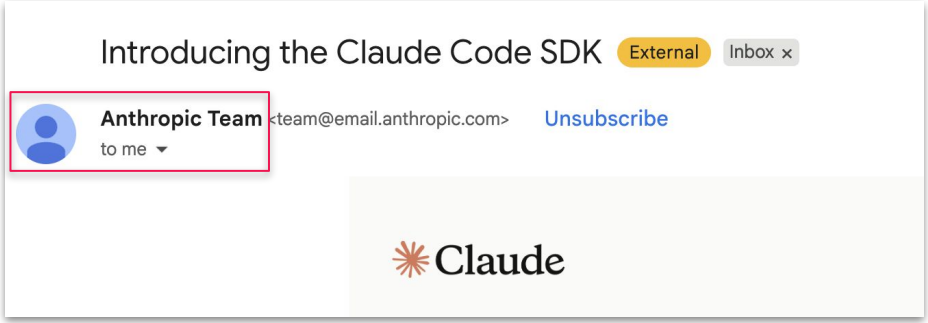
# Logo display in Email clients




**VMC Certificate**




**CMC Certificate**




**No BIMI set up**

# BIMI Certificate

**What's different in BIMI Certificate?**
**Mark type** specified in Subject
**VMC** includes trademark details (office, country, ID)
**CMC** includes a verified website URL showing prior logo use

**EKU** contains BrandIndicatorforMessageIdentification
**CertificatePolicies** include a Mark Certificate Policy identifier
**Logotype extension** embeds the SVG logo

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            59:e4:01:44:32:db:c7:03:7
        Signature Algorithm: sha384
        Issuer: (CA ID: 305221)
            commonName
            organizationName
            countryName
        Validity
            Not Before: Aug 27 09:53:39 2025 GMT
            Not After : Sep 28 09:53:38 2026 GMT
        Subject:
            1.3.6.1.4.1.53087.1.4 = 1163896
            1.3.6.1.4.1.53087.1.3 = IN
            1.3.6.1.4.1.53087.1.2 = Office of the Controller General of Patents, Designs and Trade Marks
            1.3.6.1.4.1.53087.1.13 = Registered Mark
            organizationName              = KOTAK MAHINDRA ASSET MANAGEMENT COMPANY LIMITED
            streetAddress                 = 27BKC, C27, G BLOCK, BANDRA KURLA COMPLEX, BANDRA
            localityName                  = Mumbai
            stateOrProvinceName           = Maharashtra
            countryName                   = IN
            jurisdictionStateOrProvinceName = Maharashtra
            jurisdictionCountryName       = IN
            serialNumber                  = U65991MH1994PLC080009
            businessCategory              = Private Organization
```

VMC Certificate

```
X509v3 Certificate Policies:
    Policy: 2.16.840.1.114412.0.2.5
      CPS: http://www.digicert.com/CPS
    Policy: 1.3.6.1.4.1.53087.1.1
```

```
X509v3 Extended Key Usage:
    1.3.6.1.5.5.7.3.31
```

Certificate Policies & EKU

```
Subject:
    1.3.6.1.4.1.53087.5.1 = https://web.archive.org/web/20240426120415/https://www.habitatery.com/
    1.3.6.1.4.1.53087.1.13 = Prior Use Mark
    commonName                    = HABITATERY, INC.
    organizationName              = HABITATERY, INC.
    streetAddress                 = 607 Shelby St
    localityName                  = Detroit
    stateOrProvinceName           = Michigan
    countryName                   = US
    serialNumber                  = 800633808
    businessCategory              = Private Organization
    jurisdictionStateOrProvinceName = Michigan
    jurisdictionCountryName       = US
```

CMC Certificate

```
1.3.6.1.5.5.7.1.12:
    0000 - 30 82 01 de a2 82 01 da-a0 82 01 d6 30   0..........0
    000d - 82 01 d2 30 82 01 ce 30-82 01 ca 16 0d   ...0...0.....
    001a - 69 6d 61 67 65 2f 73 76-67 2b 78 6d 6c   image/svg+xml
    0027 - 30 23 30 21 30 09 06 05-2b 0e 03 02 1a   0#0!0...+....
    0034 - 05 00 04 14 e2 8e a5 34-fc 98 74 57 a3   .......4..tW.
    0041 - 85 62 9c 4e db 76 22 bf-a2 b1 28 30 82   .b.N.v"...(0.
    004e - 01 92 16 82 01 8e 64 61-74 61 3a 69 6d   ......data:im
    005b - 61 67 65 2f 73 76 67 2b-78 6d 6c 3b 62   age/svg+xml;b
    0068 - 61 73 65 36 34 2c 48 34-73 49 41 41 41   ase64,H4sIAAA
    0075 - 41 41 41 41 41 43 6c 56-51 32 32 36 44   AAAAAClVQ226D
    0082 - 4d 41 78 39 37 31 64 59-33 6d 74 72 63   MAx971dY3mtrc
    008f - 75 46 53 4a 6b 4b 31 36-2f 4e 2b 6f 61   uFSJkK16/N+oa
    009c - 4d 70 52 47 49 4a 67 71-7a 51 76 31 2b   MpRGIJgqzQv1+
    00a9 - 41 72 74 6f 69 4b 38 66-4f 4f 63 65 32   ArtoiK8fOOce2
```

Logotype extension

Certificate centered on Brand identity, Trademark and Logo.

# VMC vs CMC

| Certificate Type | Mark Type (subject:markType value) | Description |
|---|---|---|
| **Verified Mark Certificate (VMC)** | **Registered Mark** | Mark registered and in good standing with an authorized Trademark Office. |
| **Verified Mark Certificate (VMC)** | **Government Mark** | Mark granted or claimed through official statute, regulation, or government action. |
| **Common Mark Certificate (CMC)** | **Prior Use Mark** | Common Mark validated by proof of continuous logo use on website for at least 12 months. |
| **Common Mark Certificate (CMC)** | **Modified Registered Mark** | A modified variant of an underlying Registered Mark (e.g., rearranged text or minor design removal) |

# BIMI validation

## BIMI

TXT Record

published on default._bimi.redsift.com

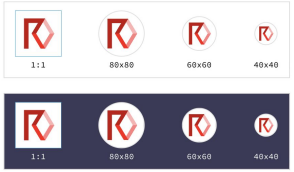
```
v=BIMI1; l=https://dynamic.bimi.red
sift.com/WSrSuiVOp/redsift.com/defa
ult/logo.svg; a=https://dynamic.bim
i.redsift.com/WSrSuiVOp/redsift.co
m/default/redsift_limited_132087795
2.pem
```

Extracted from your DNS configuration

### VMC Certificate

VMC is valid

VMC certificate from A-TAG pem-url




Valid for domains: redsift.com, redsift.co, redsift.io

Issuer: DigiCert, Inc.

Assertion Mark: Registered Mark

Expires on 29/07/2026 (286 days left)

## DMARC

TXT Record published on

_dmarc.redsift.com

```
v=DMARC1; p=reject; pct=100; sp=rej
ect; rua=mailto:6301a309@inbox.ondm
arc.com; ruf=mailto:6301a309@inbox.
ondmarc.com; adkim=r; aspf=r; fo=1;
rf=afrf; ri=3600
```

Extracted from your DNS configuration

### DMARC

DMARC passed.

**DMARC** stands for *Domain-based Message Authentication, Reporting and Conformance*.

It is a protocol that was built to enforce **the existing SPF and DKIM protocols**.

Show more

## SPF

TXT Record published on redsift.com

```
v=spf1 include:_u.redsift.com._spf.
smart.ondmarc.com ~all
```

Extracted from your DNS configuration

### SPF

SPF **authentication passed.**

SPF stands for **Sender Policy Framework**. It was developed to combat sender address forgery. It is an authentication protocol which **verifies the MAIL FROM or HELO/EHLO identities during email transmission**.

Show more

Red Sift Free BIMI Checker - https://redsift.com/tools/bimi

# BIMI support in Email clients

## Supports BIMI

- docomo
- Zoner
- CLOUDMARK
- onet POCZTA
- ZONE
- Fastmail
- GMX WEB.DE
- Google
- yahoo!
- au
- LA POSTE
- Zoho Mail
- COMCAST

## Considering BIMI

- atmail
- QUALITIA
- BT
- SEZNAM.CZ
- mail.com
- @nifty
- YAHOO! JAPAN

## Does not support BIMI

- Microsoft

Source: BIMI Working Group

# BIMI CAs (Mark Verifying Authorities)

| Name of MVA | CPS | CT Log | CRL |
|---|---|---|---|
| DigiCert | Digicert CPS | https://gorgon.ct.digicert.com/log | DigiCertVerifiedMarkIntermediateCA.crl |
| GlobalSign | GlobalSign CPS | https://gorgon.ct.digicert.com/log | gsgccr42verifiedmarkca2023.crl |
| SSL.com | SSL CPS | https://gorgon.ct.digicert.com/log | SSL.com-VMC-Root-2024-ECC.crl<br>SSL.com-VMC-Root-2024-RSA.crl<br>SSL.com-VMC-I-E1.crl<br>SSL.com-VMC-I-R1.crl |

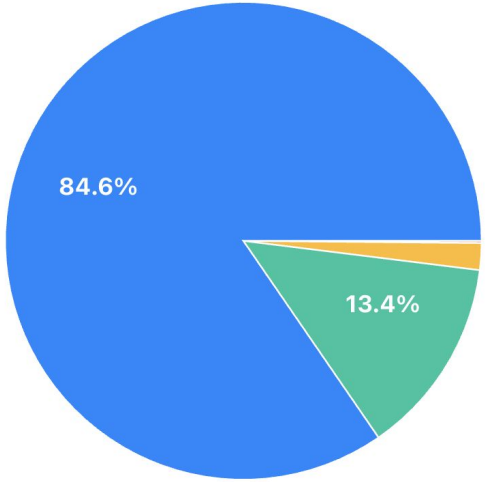
BIMI Mark Certificate Issuers: https://bimigroup.org/vmc-issuers/

Others: Entrust stopped issuance of VMC certificates effective May 12th, 2025. Their public certificate business sold to Sectigo.

# Certificate Transparency

BIMI Certificates by Certifying Authorities

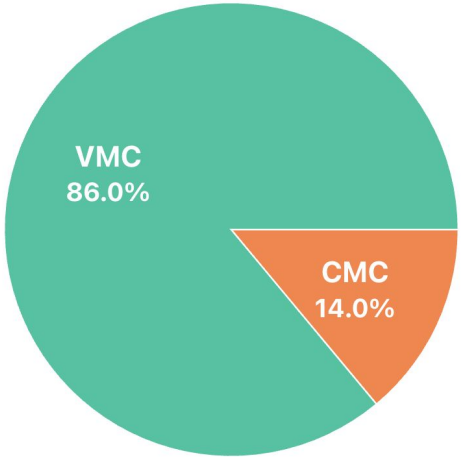
Total: 6,222 valid certificates




DigiCert: 5,264 Entrust: 836 GlobalSign: 113 SSL.com: 9

VMC vs CMC Certificate distribution

Total: 6,222 valid certificates




Verified Mark Certificate: 5,354 Common Mark Certificate: 868

DigiCert Gorgon CT log - Total entries: 26,776

Timestamp: Thursday, 16-Oct-25 08:48:43 UTC

# BIMI CT use cases

- Introduces a **new transparency dataset** - a public dataset of verified logos, trademarks, and brands.
- **Monitor** certificate issuance, discovery and expiry.
- **Audit CA** compliance: missing OIDs, invalid SVGs, non-Tiny-PS files.
- **Measure ecosystem health** and adoption across CAs.
- **Detect logo misuse** and brand impersonation (lookalikes).
- **Identify** hidden text, Unicode tricks, and mixed-script spoofing in embedded SVG.
- **Detect mismatches** between logos and certificate organization names.

# Transparency through CT log

| 🔢 Issue | ⚠️ Finding | 🧭 Impact / Why It Matters | 🧩 Instances |
|---|---|---|---|
| 1 | Missing required Mark Certificate General Policy Identifier (`1.3.6.1.4.1.53087.1.1`) | Critical CA compliance failure. Certificate violates BIMI/VMC baseline and should never be issued. | 1 |
| 2 | Missing required subject:priorUseMarkSourceURL (`1.3.6.1.4.1.53087.5.1`) | Post-April 2025 CMC rule violation. Indicates lag in CA enforcement of new BIMI standards. | 1 |
| 3 | Invalid SVG content found in logotype extension | Malformed BIMI logo breaks trust chain and mail-client rendering. | 6 |
| 4 | Unsupported SVG Tiny PS elements (`foreignObject`, `image`) | Potential security risk. Active or embedded content in what should be a safe static logo. | 9 (5 + 4) |
| 5 | SVG Invalid or missing `<title>` element | Brand identity failure. Required for accessibility and CA verification of company name. | 81 |

# Transparency through CT log

| 🔢 Issue | ⚠️ Finding | 🧭 Impact / Why It Matters | 🧩 Instances |
|---|---|---|---|
| 6 | SVG root element missing required attributes (`baseProfile`, `version`) | Spec non-compliance. Formally invalid Tiny PS SVG that may fail CA or client validation. | 88 (77 + 11) |
| 7 | SVG Invalid or missing `viewBox` values | Rendering issue. Causes inconsistent scaling across clients. | 146 (7 + 139) |
| 8 | SVG root contains forbidden `x` / `y` attributes | Minor spec violation. Could misalign logo in BIMI preview. | 33 |
| 9 | SVG file exceeds 32 KB limit | Performance or delivery concern. Not security-critical but flagged by validators. | 44 |

**Full list**: https://gist.github.com/bhushan5640/7220572da78fe7f444989d45e9ae6a65

# Policy violations

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            0a:78:36:2b:18:03:5e:d7:3e:5f:64:9b:eb:bf:24:96
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: (CA ID: 216998)
            commonName                = DigiCert Verified Mark Intermediate
            organizationalUnitName    = www.digicert.com
            organizationName          = DigiCert, Inc.
            localityName              = Lehi
            stateOrProvinceName       = Utah
            countryName               = US
        Validity (Expired)
            Not Before: Oct  4 00:00:00 2019 GMT
            Not After : Apr  1 00:00:00 2020 GMT
        Subject:
            1.3.6.1.4.1.53087.1.6 = CNN
            1.3.6.1.4.1.53087.1.4 = 5817930
            1.3.6.1.4.1.53087.1.3 = US
            organizationName          = Cable News Network, Inc.
            localityName              = Atlanta
            stateOrProvinceName       = Georgia
            countryName               = US
            postalCode                = 30303
            streetAddress             = 190 Marietta St NW
            serialNumber              = 2976730
            jurisdictionStateOrProvinceName = Delaware
            jurisdictionCountryName   = US
            businessCategory          = Private Organization
```

Missing required Mark Certificate General Policy

https://crt.sh/?q=b363c50ab320d759de51eed2003129f3c2051571624ff4ec17283f7f0b1fec5d

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            0d:bd:70:5e:82:86:fb:be:ed:ad:54:0e:e3:0d:87:30
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: (CA ID: 406116)
            commonName                = SSL.com VMC Issuing RSA CA 1
            organizationName          = SSL Corporation
            countryName               = US
        Validity
            Not Before: Apr 17 21:44:28 2025 GMT
            Not After : May 18 21:44:28 2026 GMT
        Subject:
            jurisdictionCountryName   = US
            jurisdictionStateOrProvinceName = Nevada
            streetAddress             = 3100 Richmond Ave
            businessCategory          = Private Organization
            1.3.6.1.4.1.53087.1.13 = Prior Use Mark
            serialNumber              = NV20081614243
            organizationName          = SSL.com (SSL Corp.)
            localityName              = Houston
            stateOrProvinceName       = Texas
            countryName               = US
```

Missing required subject:priorUseMarkSourceURL

https://crt.sh/?q=3cfc8bede0b4c520d345f8e7f13dba1bec1643660d0448cc6178eb69f43d8741

# SVG title not matching company name

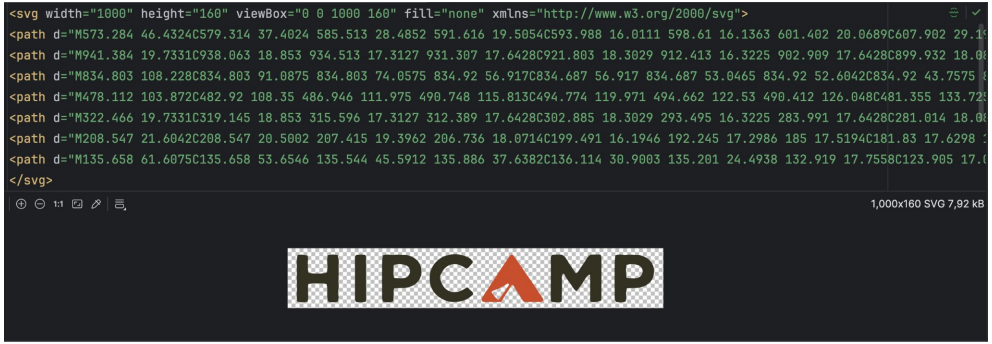

```
{
    "brandName" : "ArtPix LLC",
    "domain" : "artpix3d.com",
    "certId" : "838478519c57e355a7f8994099a85866b6985d7442c12e8a9ffbf937b76619b2",
    "issuer" : "CN=DigiCert Verified Mark RSA4096 SHA256 2021 CA1,O=DigiCert, Inc.,C=US",
    "svgTitle" : "Solarmora Services",
    "certificateType" : "VMC"
}

{
    "brandName" : "GoFundMe Inc.",
    "domain" : "gfmgivingfund.org",
    "certId" : "90790d2283c8a78ea3fd72deec86c977434af5a9c67cdf587c25e47205937f6a",
    "issuer" : "CN=DigiCert Verified Mark RSA4096 SHA256 2021 CA1,O=DigiCert, Inc.,C=US",
    "svgTitle" : "bimi-svg-tiny-12-ps",
    "certificateType" : "VMC"
}
```


**BIMI Policy**: "A <title> element must be included that reflects the company name, though there are no strict requirements for the content of the element."

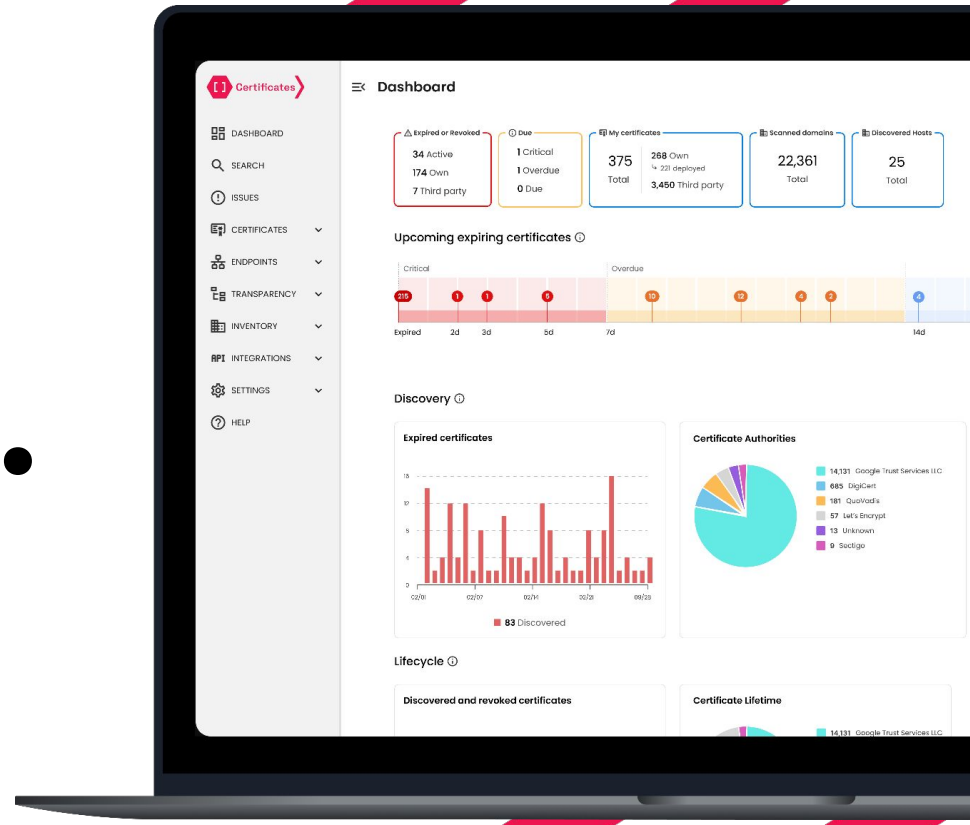
# SVG missing mandatory elements

**Certificate ID**: 0a9fdf0f7ad07da3d86c21761904b8552f2704bcaa971375832c3427da3f6783
Brand Name: Hipcamp, Inc.
Certificate Type: VMC
Not Before: Fri May 03 17:54:45 CEST 2024, Not After: Sat May 03 17:54:44 CEST 2025
Subject: 2, www.hipcamp.com
Issuer: CN=Entrust Verified Mark CA - VMC2,O=Entrust, Inc.,C=US
**Errors**:
- SVG root element missing required baseProfile="tiny-ps" attribute
- SVG root element missing required version="1.2" attribute
- SVG missing required <title> element reflecting company name



RED SIFT

# Thank you.

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